

Amimul Ehsan Zoha

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<https://amimul1.github.io/>

EDUCATION

George Mason University, Fairfax, VA | PhD, Information Technology |

Incoming Fall 2026

Advisor: **Dr. Marcos Zampieri** | Focus: Applied Natural Language Processing in Linguistics, Healthcare, and Education

The University of Arizona, Tucson, AZ | GPA 3.90 | BS and MS Computer Science

May 2026

Coursework: Algorithms for NLP, Text Retrieval & Web Search, Object-Oriented Programming & Design, Computer Programming 1-2, Machine Learning, Data Structures & Algorithms, Software Engineering, Neural Networks, Database Design, Web Development, Systems Programming & UNIX, Operating Systems, Mobile Computing.

WORK EXPERIENCE <https://amimul1.github.io/>

University of Arizona MIS Dept.

| Graduate Research Assistant, Deep Target NLP Lab |

Jan 25 – May 26

- Published and presented at **IEEE ICHI 2026** (Minneapolis, MN): "Fine-Tuning BERT with Privacy-Preserving LLM-Labeled Synthetic Training Data: An Application to Valley Fever Symptom Extraction."
- Built a privacy-preserving clinical NLP/LLM pipeline to detect Valley Fever phenotypes (symptoms + SDOH) from EHR text, curated a 1,200-sentence challenge test set with implicit symptoms, complex medical jargon, compound negations, and coordinated expert medical annotation.
- Compared trade-offs of encoder-only clinical transformers (BioClinical ModernBERT) with small quantized local LLMs (MedGemma, Llama) for deployment on institutional hardware. Advisor: **Dr Gondy Leroy**.
- Directed research under Dr. Liangming Pan on LLM logical reasoning, designing controlled experiments on step generalization, distractor robustness, negation sensitivity using FLDx2 corpus.

University of Arizona

| Teaching Assistant (6 semesters) |

Aug 2021 – May 2024

Courses: CSC 120 (Computer Programming, **Python**); CSC 210 (Software Development, **Java**); CSC 110 (**Python**)

- Led sections of 20+ students in Python and Java problem-solving, held office hours to help debug code and reinforce lecture concepts across multiple semesters.
- Evaluate student programming assignments, develop tests cases, and provide feedback on their code.
- Conducted interviews of 50+ individuals during the hiring of new TAs in the CS department and mentored students conducting feedback meetings.
- Provided career mentorship to students, sharing coursework, research, internship and interviewing experience.

UA AI Club, Tucson, AZ

| Vice President & Co-Founder |

August 24 - Present

- Co-founded and lead a 100+ member student organization. Organized faculty talks and technical workshops on LORA fine-tuning, RAG pipelines, local LLM inference, and specialized chatbot development.
- Represented the club at UArizona Hackathon and university events; led hands-on technical workshops.

AI Core, ICDI, Tucson, AZ.

| Student AI Engineer & Teaching Specialist |

May 24 -Dec 24

- Designed and delivered a 4-week instructor-led bootcamp (CustomGPTs & Web API) for 40+ interns, taught LLM fundamentals, prompt engineering, and API integration from scratch to a non-CS audience.
- Led hands-on technical workshops reaching diverse academic and professional audiences.

- Managed and developed LLM powered applications and chatbots for clients using LangChain by leveraging Retrieval Augmented Generation (RAG) and Agentic Architectures. Track and Manage API costs and interns.

American Express GBT, Chicago, IL.

| Software Development Intern |

Summer 2023

- Used Kotlin, Java and XML to add a Sustainability banner to the Android App Rail Search experience which contains detailed co2 emission info by following Egencia Design System Conventions and Figma.
- Worked with Egencia satellite service to return the co2 emission data to an API endpoint which serves both the Android and IOS app with co2 emission data using Jackson Library and REST API, tested on Postman.
- Implemented Jetpack Compose Egencia Design System Component “Callout Card Component” with custom Themes and Icons which would be reusable by Egencia Android App.

PROJECTS

GitHub: <https://github.com/amimull1>

- **Valley Fever Symptom Extractor (Python, NLTK, MedSpaCy, BioBERT, LLMs)**

- **Disease Prediction (Python, scikit-learn, KNN, Logistic Regression)**

Applied machine learning to the UCI Cleveland dataset to predict heart disease using KNN, Kernelized KNN (RBF), and Logistic Regression. Performed manual grid search and k-fold cross-validation. Evaluated models using different metrics.

- **Multi-Label Emotion Classification (Python, Keras, CNN)**

Developed a convolutional neural network for multi label emotion detection, achieving 0.83 micro-F1 on a 7-class text classification task. Applied threshold optimization, dropout regularization, and early stopping.

- **FitGuide Android App:**

Engineered a Java/XML-based fitness android app in a two-person group following Scrum principles. App uses Firebase database integration and features secure login, workout routine tracking, interactive workout guides, and meal plans.

- **Operating System Kernel:**

Devised a functional operating system kernel from scratch in C. Prototyped and developed process control and communication mechanisms and devices. Emulated OS behavior with the USLOSS library in a docker env.

- **Socialite: Full Stack Social Media App:**

Programmed an online social media website. Implemented the server using Express, and client side using HTML, CSS, and JS which makes requests to server using the **Fetch API**. Used MongoDB as database.

- **Dependency Parser (Arc-Standard based):**

Implemented a transition-based parser with an SVM-learned oracle over stack & buffer features (POS, lemma, bigrams), Filtered non-projective trees, extracted ~30 k training configurations, trained a LinearSVC classifier. Reached a 61.9% Unlabeled Attachment Score on the UD treebank.

- **Jeopardy IR QA Engine (Java, Lucene, BM25, GPT-4o-mini)**

Built a Lucene-based retriever over ~100 K Wiki pages (custom analyzer); baseline $P@1 = 0.29$, $MRR = 0.36$. Added GPT-4o-mini reranking on top-20 candidates, boosting both $P@1$ and MRR to ≈ 0.53 .

- **Log structured File System:**

Developed a FUSE (File system in the user space) on a Linux Kernel using C. The File system implements the LogStructured File System including segment cleaning.

- **DIY Resume Master (Python, Local LLM with Ollama, GPT-4o, Selenium, LaTeX)**

CORE TECHNOLOGIES AND TECHNICAL SKILLS

Programming Languages: **Java, Python, C, HTML, CSS, JavaScript, SQL, Kotlin**

Tools: HPC, PyTorch, Transformers, Docker, Pandas, Hugging Face, Unsloth, LLaMA-Factory, Tensorflow, VS Code (Copilot, Cursor), Prompt Engineering, Retrieval-Augmented Generation (RAG); Embedding k-NN few-shot, Git, LoRA, PEFT, Quantization, Fine-tuning, Weights and Biases, Synthetic Dataset, Benchmarking Evaluation, NLTK, spaCy, Scikit-learn, GitHub, AWS, LLMs, Jira, Express, Node.js, MongoDB, Trello, APIs, Postman, Scrum, React, CI/CD, Android Studio, LangChain, Figma, JavaFX, Linux, Junit, Lucene, REST APIs, Jupyter notebooks, NumPy, Matplotlib.